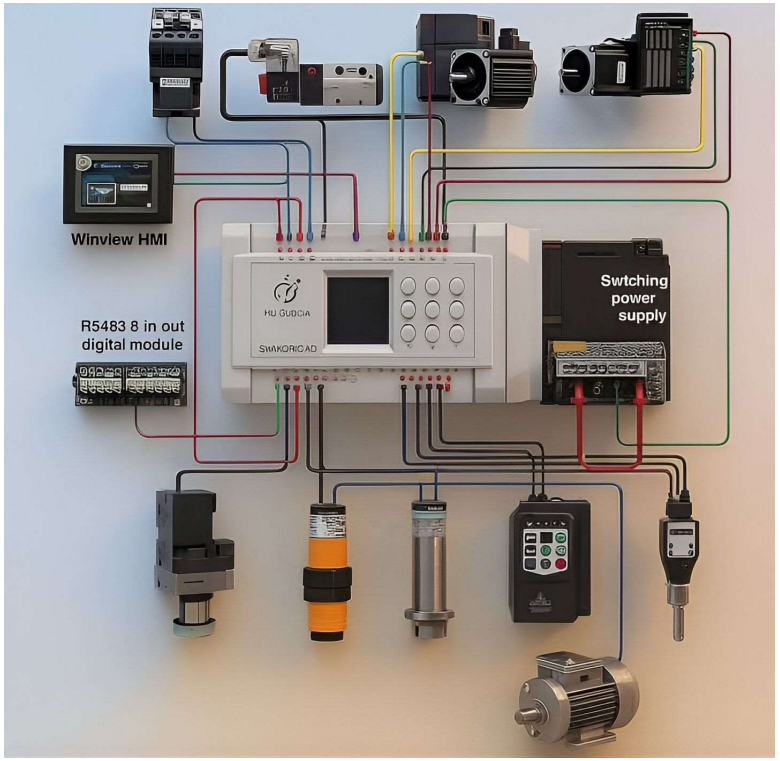
PLC Basics

Introduction to PLC and Automation

PLC

PLC = **P**rogrammable **L**ogic **C**ontroller

- An industrial digital computer for automation of electromechanical processes.
- Replaces hard-wired relays, timers, and counters.
- Can be reprogrammed easily for different control tasks.



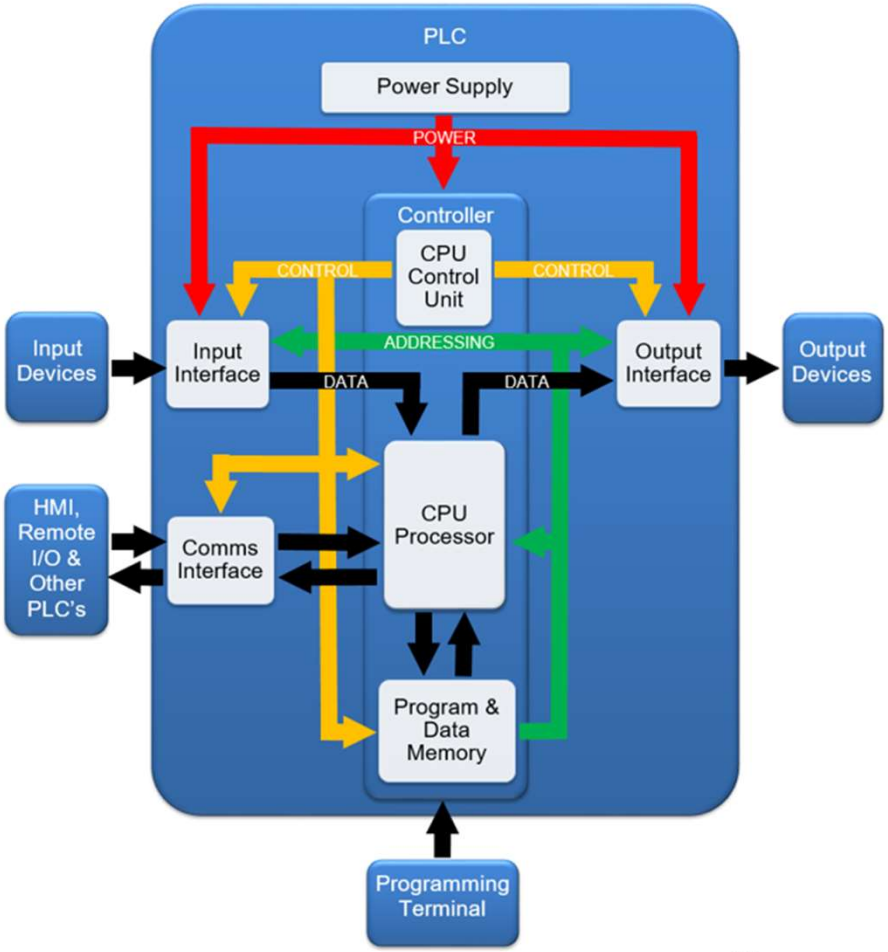
Core Components of a PLC

Power Supply:

- Converts AC to DC for internal use.
- Converts AC voltage (120/240V) into the regulated DC power (typically 24V DC or 5V DC) needed for the CPU and I/O modules.

Central Processing Unit (CPU) Module :

- **The Brain:** Executes the control program and manages system communications.
- **Memory:** Stores the operating system and the user's program/data (timers, counters, I/O status).



Core Components of a PLC : I/O Modules

Input/Output (I/O) Modules: Input/Output (I/O) Modules are responsible for communication between the CPU and the external devices

- **Input Module:** Input modules are responsible for detecting the status of input signals, such as push-buttons, switches, temperature sensors, and other devices.
- **Output Module:** Output modules are responsible for sending signals to output devices, such as motors, valves, and other devices. They convert the signals from the processor into a format that the devices can understand.

Network I/O Modules:

- Network I/O modules are designed to handle signals that are transmitted over a network, such as Ethernet, Modbus, Profibus, etc.
- Facilitates high-speed data exchange with other controllers, HMIs, and SCADA systems using industrial protocols.

Types of I/O Modules

Discrete I/O Modules:

- Discrete I/O modules are designed to handle digital signals that are either on or off.

Analog I/O Modules:

- Analog I/O modules are designed to handle signals that vary continuously over time, such as temperature, pressure, and flow rate.

Fixed and Modular PLC Architecture

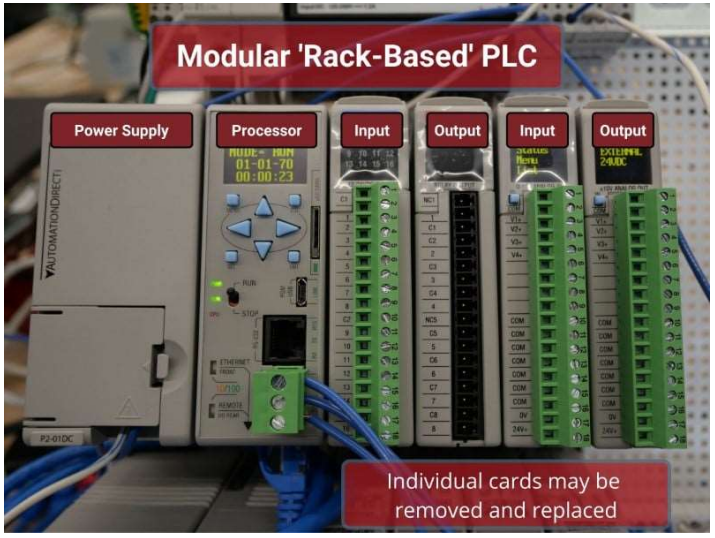
Fixed PLC Type Architecture:

- A single unit architecture where all the hardware components are embedded into a single unit.
- Small in size, quick and easy to mount, and low in cost, they provide an economical solution for basic applications.



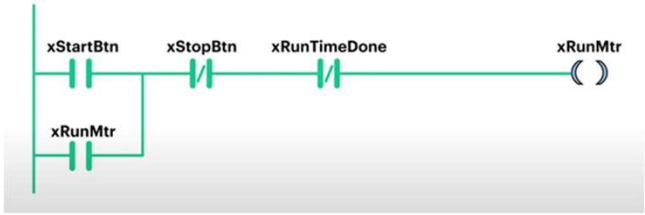
Modular PLC Type Architecture:

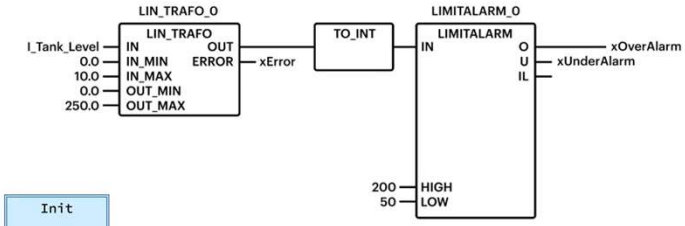
- A modular PLC has a separate module for each of its hardware components. Each PLC module is interconnected using a common mounting system.
- A modular PLC can be configured to be application specific.

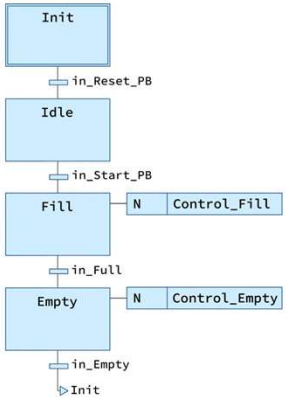


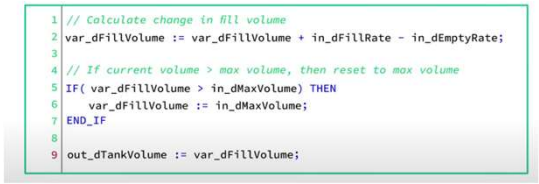
PLC Programming

1. **Ladder Diagram (LD)**
   - Graphical method resembling electrical relay logic with contacts and coils.
   - Most intuitive for technicians and used for discrete control (e.g., motor start/stop).
2. **Function Block Diagram (FBD)**
   - Uses interconnected blocks (AND, OR, timers, counters) to represent logic flow.
   - Suitable for continuous or process control and signal processing.
3. **Sequential Function Charts (SFC)**
   - Flowchart-style representation showing steps and transitions in a process.
   - Ideal for sequential operations like machine cycles or batch processes.
4. **Structured Text (ST)**
   - High-level, text-based language similar to Pascal or C.
   - Best for complex computations, loops, and conditional operations.
5. **Instruction List (IL)**
   - Low-level, assembly-like language with compact instructions.
   - Fast execution but less visual; now largely replaced by Structured Text.
   - Obsolete Programming Language









```
// Calculate change in fill volume
var_dFillVolume := var_dFillVolume + in_dFillRate - in_dEmptyRate;

// If current volume > max volume, then reset to max volume
IF( var_dFillVolume > in_dMaxVolume) THEN
    var_dFillVolume := in_dMaxVolume;
END_IF

out_dTankVolume := var_dFillVolume;
```

PLC and SCADA

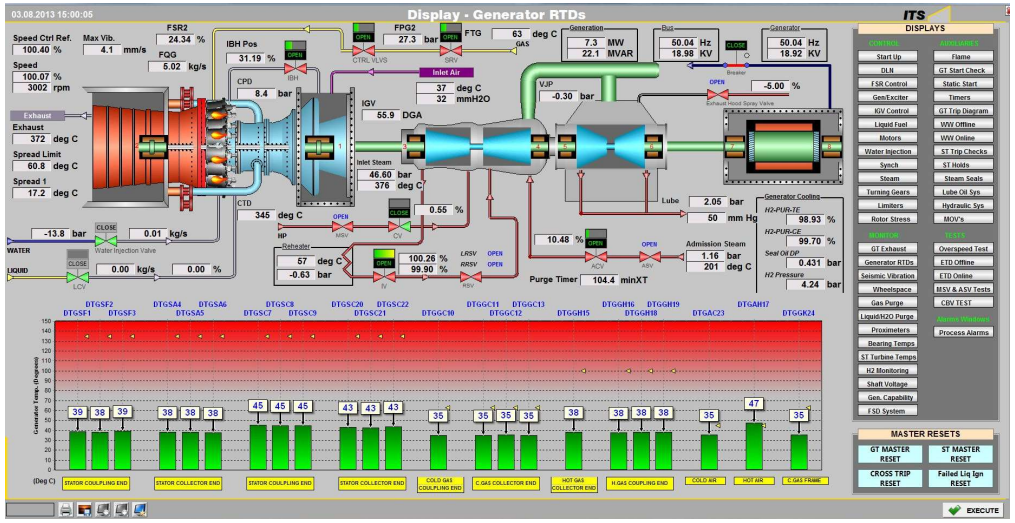
PLC:

- A PLC is a hardware controller used to automate machinery or processes at the field level.
- PLCs are real-time, local controllers , they directly control motors, valves, and lights on the factory floor.

SCADA (Supervisory Control and Data Acquisition):

- SCADA is a software-based system used for **monitoring**, **visualization,** and **supervisory control** of industrial processes.
- It connects to multiple PLCs or controllers, gathers data (e.g., temperature, flow rate), displays it on graphical screens (HMIs), and stores it in databases.
- Operators use SCADA to monitor performance, adjust setpoints, or analyze trends remotely.

e.g.: A SCADA system shows real-time data from several PLCs controlling pumps in a water treatment plant and allows operators to change settings from a control room



PLCs and Industrial Automation

- **Ruggedness**
  - Built-in isolation and robust components to withstand extreme heat, vibration, and electrical noise (EMI/RFI).
- **Deterministic Operation**
  - Executes the control program in a reliable, predictable, and repeatable **scan cycle** (real-time system). Timing is guaranteed.
- **Programming Language**
  - Uses industry-standard languages (IEC 61131-3), especially **Ladder Logic**, which is graphically intuitive for electrical technicians
- **I/O Integration**
  - Features high-voltage/current I/O with built-in isolation and signal conditioning, allowing direct connection to industrial sensors and motors.
  - **Interfacing:** Communicate with sensors, actuators, and HMIs.
  - Scalability: Can expand with extra I/O modules.

Simple PLC Circuit using Ladder Program

